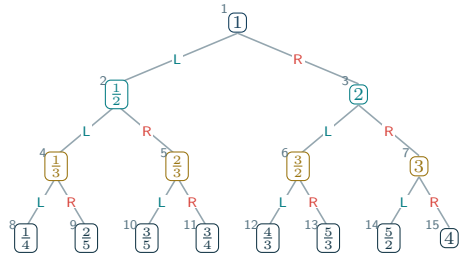
COMBINATORICS ON ONE PAGE

A tree of fractions as a growing adjacency matrix

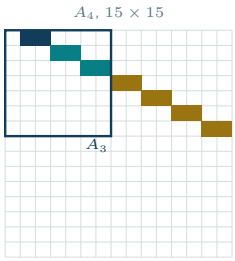
Two unary maps label a binary tree, its matrix grows from N to $2N+1$, and the weight $1/(pq)$ is conserved on every level.

1 / THE RULE AND THE PICTURE

A binary tree of fractions, with the adjacency matrix recording only its shape



Stern–Brocot labels, heap index i in grey; L to $2i$, R to $2i+1$



rows i, columns j; new ones gold

Node i has children $2i$ and $2i+1$, so the matrix has $A[i][2i] = A[i][2i+1] = 1$, and one more level takes N nodes to $2N+1$. The nodes carry the fractions $f = \frac{p}{q}$ through two unary maps, the inversion J that swaps them, and the weight:

$$L(f) = \tfrac{p}{p+q}, \quad R(f) = \tfrac{p+q}{q}$$
$$J(f) = \tfrac{q}{p}, \quad w(f) = \tfrac{1}{pq}$$

and $J \circ R = L \circ J$.

2 / TWO LABELLINGS OF ONE MATRIX

Parent values give Calkin–Wilf, and reading each path backwards gives Stern–Brocot

The matrix is the same for every complete binary tree, so only the labels differ. Applying the maps to the parent value labels node i by $S_k(\cdots S_1(1))$ along its path, the Calkin–Wilf tree. The Stern–Brocot tree has the same fractions on every level, in increasing order, and puts $S_1(S_2(\cdots S_k(1)))$ at node i, which is the path read backwards, or equivalently the mediant of the two nearest ancestors:

heap index i	8	9	10	11	12	13	14	15
apply L, R to the parent value (Calkin–Wilf)	$\frac{1}{4}$	$\frac{4}{3}$	$\frac{3}{5}$	$\frac{5}{2}$	$\frac{2}{5}$	$\frac{5}{3}$	$\frac{3}{4}$	4
Stern–Brocot, increasing along the level	$\frac{1}{4}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{3}{4}$	$\frac{4}{3}$	$\frac{5}{3}$	$\frac{5}{2}$	4

For example node 11 has path L, R, R and holds $\frac{5}{2}$ in one tree and $\frac{3}{4}$ in the other.

3 / HOW THE MATRIX GROWS

A_{n+1} contains A_n and adds ones only in the rows of the old leaves

With $N_n = 2^n - 1$ nodes, $A_{n+1} = \begin{pmatrix} A_n & E \\ 0 & 0 \end{pmatrix}$, where E has the ones $(i, 2i)$ and $(i, 2i+1)$ for the 2^{n-1} old leaves, as the gold block of A_4 shows. The matrix is strictly upper triangular, $(A^k)[1][j] = 1$ exactly when node j is on level $k+1$, and $A_n^{n-1} \neq 0 = A_n^n$. The first row of $(I - A)^{-1}$ is all ones, since every node descends from the root.

levels n	nodes N	next size $2N+1$	edges $N-1$	new nodes 2^{n-1}	nilpotent
1	1	3	0	1	$A^1 = 0$
2	3	7	2	2	$A^2 = 0$
3	7	15	6	4	$A^3 = 0$
4	15	31	14	8	$A^4 = 0$
5	31	63	30	16	$A^5 = 0$
6	63	127	62	32	$A^6 = 0$

4 / THE INVARIANTS

Mass is conserved down the tree, and inversion is a mirror image

Splitting a node gives $\frac{1}{p(p+q)} + \frac{1}{(p+q)q} = \frac{1}{pq}$, so in the Calkin–Wilf labelling $w(f) = w(Lf) + w(Rf)$, which in matrix form is $Aw = w$ on the internal rows and 0 on the leaves, and every level has total weight 1: 1; $\frac{1}{2} + \frac{1}{2}$; $\frac{1}{3} + \frac{1}{6} + \frac{1}{6} + \frac{1}{3}$. Stern–Brocot levels hold the same fractions, so their sums are also 1, but there it is the Farey interval $\frac{1}{bd} = \frac{1}{b(b+d)} + \frac{1}{(b+d)d}$ that splits, so a node's weight is not sum of its heap children. Because J swaps L and R, the reciprocal of node i on level n sits at mirror index $3 \cdot 2^{n-1} - 1 - i$ in both labellings, so $w(f) = w(f^{-1})$.

5 / THE CONSTRUCTION IN FOUR STEPS

What each step does and what it checks

step	construction	check and remark
1	start at the root: $i = 1$, $f = 1$, $w = 1$, $A_1 = [0]$	one node and no edges
2	add a level: $N \to 2N+1$; L leads to $2i$ and R to $2i+1$	parent values give Calkin–Wilf labels; Stern–Brocot composes the maps with the first step outermost
3	fill the matrix: $A[i][2i] = A[i][2i+1] = 1$	the same matrix for both labellings
4	check $\sum w = 1$ and $w(f) = w(f^{-1})$	the sum is over each level; the reciprocal sits at mirror index $3 \cdot 2^{n-1} - 1 - i$

ASIDE / TRY THESE

Check that level 5 has 16 fractions with weights summing to 1; index 13 (path R, L, R) holds $\frac{5}{3}$ in both trees.

LINKS TO OTHER SHEETS

The mediants sheet and the ternary-tree sheet: the mediant is the Stern–Brocot step.

Convention. L, R unary maps; J inversion; $w = 1/(pq)$; heap index i, children $2i, 2i+1$; all claims checked with exact fractions.

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